

Availability Assessment of Dual-frequency GNSS-based Augmentation Systems under Equatorial Ionospheric Scintillations

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Abstract

This study investigates a wide range of equatorial scintillations to evaluate their impacts on the availability of dual-frequency Global Navigation Satellite System (GNSS)-based augmentation systems. A two-component power-law phase screen model is used to implement scintillation simulations for all possible sets of phase screen parameters within their empirical range. Two types of scintillation effects, phase transition and loss of lock, are conservatively quantified in terms of navigation availability. This study specifies equatorial scintillation conditions with the scintillation indicators, S_4 and τ_0 , to search for the worst scintillation impact among all simulation results assigned to each grid of S_4 and τ_0 . Aviation availability of a GPS/Galileo dual-frequency advanced receiver autonomous integrity monitoring (ARAIM) is assessed for the scintillation scenario selected from the real scintillation data. The simulation results show that the availability is highly dependent on the reacquisition time and the C/N_0 threshold which are closely related to the receiver design specifications and user settings. The proposed approach allows availability assessment of GNSS-based augmentation systems for any scintillation conditions specified by indicators $\{S_4, \tau_0\}$ where the indicators can be obtained in near real-time.

1. INTRODUCTION

The equatorial region is well known to be the most challenging region for the operation of GNSS-based augmentation systems due to the impact of ionospheric scintillation. One of the major sources of equatorial scintillation is the irregular structure in the F region of the ionosphere, called the equatorial plasma bubble (EPB). The EPB has scale sizes down to tens of meters that cause diffractive scattering of the radio waves (Kintner et al., 2007). The frequency scaling of the carrier phase is violated in the presence of diffractive fluctuations during equatorial scintillation, which means that the diffractive effects of equatorial scintillation cannot be eliminated with the ionosphere-free combination of dual-frequency signals (Nguyen et al., 2019). Therefore, equatorial scintillation can induce significant threats to GNSS-based augmentation systems even in their dual-frequency operations.

The diffractive scintillation mostly accompanies deep signal fades along with abrupt phase changes, so-called “canonical fading,” which can break the linear regime of the phase lock loop (PLL) and leads to the following effects on phase tracking in GNSS receivers: phase transition and loss of lock (Humphreys et al., 2009). Phase transitions introduce an integer-cycle bias into the phase measurements (Breitsch et al., 2020) and a step input to the carrier-smoothing filter to cause additional errors which may lead to a significant integrity threat to aviation users. If the phase errors continue to exceed a certain range of the PLL, then loss of lock (so-called frequency unlock in Humphreys et al., 2010) may occur and lead to initialization of the carrier-smoothing filter and complete signal/satellite loss to cause integrity and continuity threats, respectively (Lee et al., 2017; Seo et al., 2011).

Our previous work in Sun et al. (2021) assessed the impact of equatorial scintillation on the availability of dual-frequency GNSS-based augmentation systems by utilizing a stochastic model to simulate realistic behaviors of fading processes during equatorial scintillation. The model simplifies the correlated fading process in dual-frequency signals as a Markov process with statistical parameters extracted from real scintillation data. However, the effect of phase transitions is not accounted for in Sun et al. (2021). Also, the occurrence of loss-of-lock is declared with a fixed threshold of -10 dB for deep signal fading where the criteria of the loss-of-lock depends on the tracking capability or receiver settings for commercial receivers (Forte, 2012). The proposed data-based model is unable to simulate all possible scintillation conditions (e.g., structural composition of irregularities and propagation geometry); hence, all possible effects of equatorial scintillations are not fully accounted for in Sun et al. (2021).

A two-component power-law phase screen model presented in Carrano & Rino (2016) is a physics-based model which can describe all possible scintillation conditions specified by five phase screen parameters $\{U, p_1, p_2, \mu_0, \rho_F/v_{eff}\}$ that are responsible for the scintillation observed on a receiver. Xu et al. (2020) proposed a scintillation simulator based on the two-component power-law phase screen model which requires only two scintillation indicators: the amplitude scintillation index S_4 and the decorrelation time τ_0 , which is available on board in near real-time for aviation users; hence, it can specify most typical scintillation effects with only these two indicators. Breitsch et al. (2020) utilizes this simulator to generate scintillations under a variety of equatorial scintillation scenarios specified by S_4 , τ_0 and baseline C/N_0 , and investigates the cycle slip occurrence rate corresponding to the relevant scenarios.

GNSS-based augmentation systems provide a safety-critical aviation service which is required to consider all possible scintillation-induced threats to aviation users to meet its high level of integrity and continuity. We can overcome the limitation of data-based modeling of the scintillation effects in Sun et al. (2021) by simulating a wide range of equatorial scintillation with varying statistical characteristics via the simulator proposed in Xu et al. (2020). The simulation method assumes a one-to-one numerical mapping between a set of scintillation indicators $\{S_4, \tau_0\}$ and phase screen parameters $\{U, p_1, p_2, \mu_0, \rho_F/v_{eff}\}$. However, there exists statistical uncertainty in S_4 and τ_0 with a finite time window (normally 60 s) (Sun et al. 2023); hence, different scintillation impacts resulting from different combinations of the phase screen parameters can be assigned to a given set of scintillation indicators $\{S_4, \tau_0\}$. Therefore, an exhaustive search for the worst scintillation impacts is required to cover all possible scintillation-induced threats for each set of scintillation indicators $\{S_4, \tau_0\}$.

The purpose of this study is to quantify the worst scintillation impacts for each set of indicators $\{S_4, \tau_0\}$, and assess the availability of dual-frequency GNSS-based augmentation system under a scintillation scenario of $\{S_4, \tau_0\}$. Section 2 introduces a two-component power-law phase screen model and scintillation simulations for this study. Section 3 presents a conservative approach to quantify two scintillation effects: phase transition and loss of lock in terms of navigation performance of GNSS-based augmentation systems. Section 4 provides the results of the worst scintillation impacts according to the scintillation indicators $\{S_4, \tau_0\}$. Section 5 presents the availability simulation for a selected equatorial scintillation scenario, and conclusions are summarized in Section 6.

2. PHASE SCREEN SIMULATION OF SCINTILLATION

2.1. Two-component Power-law Phase Screen Model

A statistical characterization of scintillation is available by employing the phase screen approximation, which replaces the interaction of the radio wave with an irregularity medium by an equivalent phase-changing screen (Rino, 2011). Previous studies on the phase screen theory have developed based on a power-law spectrum model that characterizes the spectrum of a given medium structure as an inverse power-law form (Rino, 1979). Carrano and Rino (2016) have extended the power-law-based phase screen theory to support two-component spectra which allows statistically consistent representations of strong equatorial scintillation data (Rino et al., 2016). The two-component power-law model for the phase spectral density function (SDF) corresponding to the phase screen theory is defined as (Carrano & Rino, 2016):

$$\Phi_\phi(q) = C_p \begin{cases} q^{-p_1}, & q \leq q_0 \\ q_0^{p_2-p_1} q^{-p_2}, & q > q_0 \end{cases} \quad (1)$$

where C_p is the strength of turbulence, p_1 and p_2 are the phase spectral indices and q_0 is the break wavenumber which indicates inner or outer scale of the irregularity structure. Since irregularity structures in the equatorial ionosphere (i.e., EPBs) are highly elongated along magnetic field lines, the phase structure can be simplified to the corresponding one-dimensional SDF, where q is the wavenumber of the isotropic coordinate system converted from anisotropic coordinates of field-aligned structures (Rino, 1979; Rino, 2011).

The SDF can be simplified by scaling the wavenumber q with the Fresnel scale ρ_F to a normalized (and thus non-dimensional) unit such that $\mu = q\rho_F$ and $\mu_0 = q_0\rho_F$. Hence, the phase spectrum can be expressed in terms of the normalized wavenumber μ as (Carrano & Rino, 2016):

$$P(\mu) = \Phi_\phi(q)/\rho_F = \begin{cases} U_1 |\mu|^{-p_1}, & \mu \leq \mu_0 \\ U_2 |\mu|^{-p_2}, & \mu > \mu_0 \end{cases} \quad (2)$$

where the universal scattering strength U (i.e., the normalized phase spectral power at the Fresnel scale ρ_F) can be defined as (Carrano & Rino, 2016):

$$U = \begin{cases} U_1 = C_p \rho_F^{p_1-1}, & \mu_0 \geq 1 \\ U_2 = C_p q_0^{p_2-p_1} \rho_F^{p_2-1}, & \mu_0 < 1 \end{cases} \quad (3)$$

Based on the dimensionless power-law form of the phase spectrum in Equation (2), the intensity spectrum can also be expressed explicitly in terms of the four spectral parameters $\{U, p_1, p_2, \mu_0\}$ as (Carrano & Rino, 2016):

$$I(\mu; U, p_1, p_2, \mu_0) = 2 \int_0^\infty \exp\{-\gamma(\eta, \mu; U, p_1, p_2, \mu_0)\} \cos(\eta\mu) d\eta \quad (4)$$

where γ is the so-called structure interaction function (Carrano & Rino, 2016):

$$\gamma(\eta, \mu; U, p_1, p_2, \mu_0) = 16 \int_0^\infty \exp\{-P(\chi; U, p_1, p_2, \mu_0)\} \sin^2(\chi\eta/2) \sin^2(\chi\mu/2) d\chi/2\pi \quad (5)$$

Scintillation observations from GNSS signals are uniformly sampled in the time domain, whereas the phase and intensity spectra in Equations (2) and (4) are represented in the wavenumber domain. For the representation of scintillation time series with the phase screen model, the scintillation spectrum in the spatial domain is required to be translated into the temporal domain. The conversion of the normalized spatial wavenumber (μ) into the temporal Doppler frequency (f_D) is expressed as (Xu et al., 2020):

$$f_D = \frac{\mu}{2\pi \rho_F / v_{\text{eff}}} \quad (6)$$

where the Fresnel scale ρ_F determines the propagation distance from the phase screen where the irregularity structures are summarized, and the effective scan velocity v_{eff} is the rate of effective line-of-sight scan across the field-aligned, anisotropy irregularity structures (Rino, 1979; Rino, 2011). The scaling factor ρ_F/v_{eff} determines to what extent the spatial variations of the phase or intensity spectra would be projected over the temporal domain of the scintillation fluctuations. According to the space-to-time conversion in Equation (6), the intensity SDF $I(\mu)$ (or phase SDF $P(\mu)$) in the spatial wavenumber domain (in μ) can be converted to their counterparts $\Phi_I(f_D)$ (or $\Phi_P(f_D)$) in the temporal frequency domain (in f_D) as (Xu et al., 2020):

$$\Phi_I(f_D) = I\left(\frac{\mu}{2\pi \rho_F / v_{\text{eff}}}\right) 2\pi \rho_F / v_{\text{eff}} \quad (7)$$

Therefore, the five phase screen parameters $\{U, p_1, p_2, \mu_0, \rho_F/v_{\text{eff}}\}$ can specify most of the scintillations generated by different combinations of perturbation strength (i.e., U), spectral characteristics of irregularities (i.e., p_1, p_2 , and μ_0) and the propagation geometry (i.e., ρ_F/v_{eff}), based on the solutions for two-component spectra (Carrano & Rino, 2016).

2.2. Phase Screen Realization and Wave Propagation

Based on the phase screen model characterized by a one-dimensional, two-component power-law SDF in Equation (2), statistically equivalent realizations of phase screen can be generated by imposing the desired phase SDF on white noise on the discrete Fourier transform (DFT) with the length N as (Rino et al., 2018):

$$\bar{\phi}_m = \sum_{n=0}^{N-1} \sqrt{P(n\Delta\mu) \Delta\mu / (2\pi)} \eta_n e^{-2\pi i n m / N} \quad (8)$$

where η_n is a standard complex Gaussian random process with the white noise property whose variates are uncorrelated with unit-variance as $\langle \eta_m \eta_{m'}^* \rangle = \delta(m-m')$ and the Hermitian property $\eta_{N-m} = \eta_m^*$ which enables purely real realizations of phase screen (Rino, 2011).

Applying the phase screen realization $\bar{\phi}_m$, the propagation of the complex field at the Fresnel scale ρ_F (i.e., the effective propagation distance from the phase screen) is implemented through the following split-step algorithm of the forward propagation equation following the parabolic approximation (Rino, 2011; Rino et al., 2018):

$$\hat{\psi}(0; n\Delta\mu) = \sum_{m=0}^{N-1} e^{i\bar{\phi}_m} e^{-2\pi i n m / N} \quad (9)$$

$$\psi(\rho_F; m\Delta y / \rho_F) = \frac{1}{N} \sum_{n=0}^{N-1} \hat{\psi}(0; n\Delta\mu) e^{i(n\Delta\mu)^2 / 2} e^{2\pi i n m / N} \quad (10)$$

where y is a distance directed perpendicular to the propagation plane at a given height, and the complex wave field ψ can be converted from space to time domain using the effective scan velocity such that $y = v_{\text{eff}} t$.

The resulting phase ϕ and signal amplitude I are then obtained from the propagated complex wave field as

$$\phi = \text{unwrap}\{\text{atan } 2(\Re(\psi), \Im(\psi))\} \quad (11)$$

$$I = |\psi|^2 = \psi \psi^* \quad (12)$$

We extract the diffractive term of phase by subtracting the initiating phase screen (i.e., refractive term) from the phase (Breitsch et al., 2020).

2.3. Scintillation Simulation

This study simulates full combinations of five phase screen parameters $\{U, p_1, p_2, \mu_0, \rho_F/v_{\text{eff}}\}$ in their empirical range for equatorial scintillations to investigate all possible scintillation impacts on GNSS-based augmentation systems. Table 1 provides the range and step size for each parameter used for this study to implement scintillation simulations based on the two-component power-law phase screen model shown in Sections 2.1 and 2.2. The range of the phase screen parameters $\{U, p_1, p_2, \mu_0, \rho_F/v_{\text{eff}}\}$ is based on the data analysis in Xu et al. (2020) which extracted the phase screen parameters from real scintillation data segments using the irregularity parameter estimation (IPE) method developed in Carrano et al. (2012). Note that we used the Fresnel frequency $f_F (= v_{\text{eff}}/\rho_F)$ instead of ρ_F/v_{eff} because v_{eff} (the effective velocity of the satellite scan with respect to the irregularities) is typically more variable than ρ_F (the effective distance from the irregularities to the ground receiver). For this reason, the Fresnel frequency f_F is more uniformly distributed than ρ_F/v_{eff} in the real scintillation data; hence, it is more convenient to use f_F for parametric scintillation simulations.

Table 1. The range and step size of parameters used for scintillation simulations in this study

Parameter	Range	Step
U	0 ~ 3	0.3
p_1	1.5 ~ 3	0.5
p_2	2.5 ~ 4.5	0.5
μ_0	0.2 ~ 1	0.2
$f_F (= v_{\text{eff}}/\rho_F)$	0.3 ~ 3	0.3
C/N_0	30 ~ 50 (dB-Hz)	5 (dB-Hz)

In the real GNSS observations, scintillation effects on the receiver are largely dependent on the baseline noise in the signal. We consider a baseline carrier-to-noise density ratio C/N_0 as the relevant parameter determining the impact of thermal noise on the receiver. For a given C/N_0 , the noise imposed on the phase measurements can be derived as

$$n = \sigma\eta \quad \text{where} \quad \sigma = \sqrt{\frac{B}{C/N_0}} \quad (13)$$

where η is a standard complex Gaussian random variable and σ is a noise standard deviation. The noise bandwidth B is assumed to be $1/T$ where T is the integration time (i.e., sampling interval of raw measurements which is 10 ms for this study) (Breitsch et al., 2020). In this study, we consider baseline C/N_0 from 30 to 50 dB-Hz with a step size of 5 dB-Hz for scintillation simulations as shown in Table 1. Then the noise can be induced in the measurements as

$$\phi = \text{unwrap} \left\{ \text{atan} 2(\Re(\psi + n), \Im(\psi + n)) \right\} \quad (14)$$

$$I = |\psi + n|^2 \quad (15)$$

We generate over 10 hours of scintillation with a sampling interval of 10 ms for each set of modeling parameters $\{U, p_1, p_2, \mu_0, f_F\}$ and a baseline C/N_0 . A large volume of simulations is required to obtain a statistically significant measure of scintillation effects, specifically phase transition and loss of lock which will be presented in Section 3.

To investigate the scintillation impacts on dual-frequency signals, we generate scintillations of L5 in addition to those of L1 where the phase screen for L5 is scaled from L1 using the following equations for the same white noise realization in Equation (8) (Xu et al., 2020):

$$U(f_{L5}) = \begin{cases} U(f_{L1}) (f_{L1}/f_{L5})^{\frac{p_1+3}{2}} & \text{if } \mu_0(f_{L1}) \geq 1, \mu_0(f_{L5}) \geq 1 \\ U(f_{L1}) \{\mu_0(f_{L1})\}^{p_1-p_2} (f_{L1}/f_{L5})^{\frac{p_1+3}{2}} & \text{if } \mu_0(f_{L1}) < 1, \mu_0(f_{L5}) \geq 1 \\ U(f_{L1}) \{\mu_0(f_{L5})\}^{p_2-p_1} (f_{L1}/f_{L5})^{\frac{p_1+3}{2}} & \text{if } \mu_0(f_{L1}) \geq 1, \mu_0(f_{L5}) < 1 \\ U(f_{L1}) (f_{L1}/f_{L5})^{\frac{p_2+3}{2}} & \text{if } \mu_0(f_{L1}) < 1, \mu_0(f_{L5}) < 1 \end{cases} \quad (16)$$

$$\mu_0(f_{L5}) = \mu_0(f_{L1}) \sqrt{f_{L1}/f_{L5}} \quad (17)$$

$$\rho_F/v_{\text{eff}}(f_{L5}) = \rho_F/v_{\text{eff}}(f_{L1}) \sqrt{f_{L1}/f_{L5}} \quad (18)$$

where p_1 and p_2 remain the same for different frequencies. f_{L1} is the carrier frequency for GPS L1 (1575.42 MHz) and f_{L5} is for GPS L5 (1176.45 MHz).

3. QUANTIFYING SCINTILLATION EFFECTS

3.1. Phase Transition

Phase transition is a full-cycle change in phase (half-cycle for Costas discriminators) that mostly coincides with deep signal fades and abrupt phase changes referred to as canonical fading (Breitsch et al., 2020; Humphreys et al., 2009). The phase transition occurs when the diffractive field phasor circuits around or jumps across the complex origin of the discriminator (Breitsch et al., 2020; Carrano et al., 2013). The phase change with a full cycle is hard to be detected in the presence of phase variations due to receiver-satellite dynamics, oscillator noise, and total electron content (TEC) variations; hence, phase transitions can be accumulated without detection (until a loss of lock occurs) to introduce integer-cycle bias into the phase measurements.

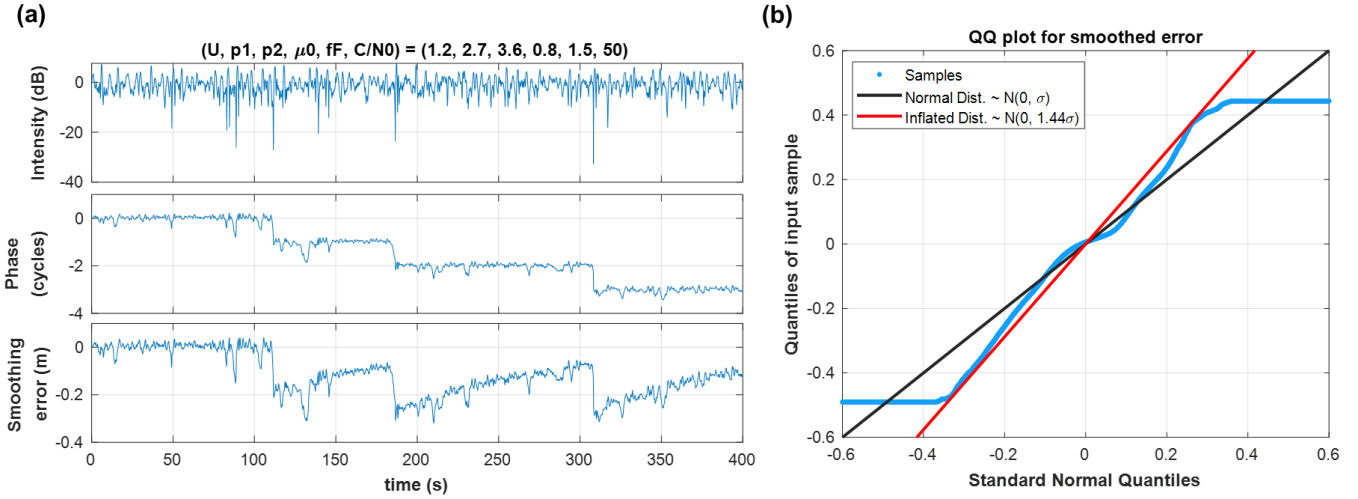


Figure 1 (a) An example of signal intensity, diffractive phase, and the resulting response of the carrier smoothing filter for a scintillation simulation with a parameter set of $\{U, p_1, p_2, \mu_0, f_F, C/N_0\} = \{1.2, 2.7, 3.6, 0.8, 0.6, 50 \text{ dB-Hz}\}$. (b) A Q-Q plot for the smoothing error induced by diffractive phase including phase transitions (blue dots) and the corresponding normal (black line) and the overbounding normal distribution (red line)

Figure 1(a) shows an example of simulated signal intensity and phase time series for a set of parameters $\{U, p_1, p_2, \mu_0, f_F, C/N_0\} = \{1.2, 2.7, 3.6, 0.8, 0.6, 50 \text{ dB-Hz}\}$. The occurrence of phase transitions (shown in the middle panel) coincides with deep signal fading (shown in the top panel). The bottom panel shows the output of the carrier smoothing filter with a 100-s time constant for the diffractive phase input. The response for the full-cycle step input due to phase transitions can introduce additional error to the carrier smoothing filter as shown in the bottom panel. When successive phase transitions occur, the response of the smoothing filter may be accumulated to cause significant smoothing error for aviation users.

In order to obtain a conservative measure of the phase transition effects, we use cumulative distribution function (CDF) overbounding to obtain a σ_{scin} overbound that covers the smoothing error distribution from the simulated diffractive phase for a given parameter set $\{U, p_1, p_2, \mu_0, f_F, C/N_0\}$ with a confidence level of $1-10^{-7}$ (DeCleene, 2000). Figure 1(b) shows a Q-Q plot for the smoothing error from the diffractive phase simulated with a set of parameters $\{U, p_1, p_2, \mu_0, f_F, C/N_0\} = \{1.2, 2.7, 3.6, 0.8, 0.6, 50\}$. Note that the CDF overbounding is achieved if the inflated Gaussian distribution is below the sample distribution for all quantiles below zero and above the sample distribution for all quantiles above zero. This indicates that the normalized smoothing error distribution (blue dots) has non-Gaussian tails violating the 1-CDF of the standard normal distribution (black line), while the distribution inflated with 1.44 overbounds the actual distribution by covering its non-Gaussian tails. In this study, we obtain σ_{scin} overbound for each L1/L5 single-frequency and ionosphere-free smoothing mode. Note that we use the smoothing filter with a 100-s time constant as specified in the dual-frequency multi-constellation (DFMC) satellite-based augmentation system (SBAS) minimum operational performance standards (MOPS) (EUROCAE, 2021).

3.2. Loss of lock

Loss of lock occurs when the phase error exceeds a certain threshold to cause frequency unlock of the PLL (Humphreys et al., 2010). Mostly, loss of lock accompanies a large phase jump after reacquisition (i.e., cycle slip), so the receiver needs to re-initialize the smoothing filter for the tracking channel. Frequent occurrence of loss of lock causes frequent initialization of the smoothing filter, which shortens the effective smoothing time leading to higher noise levels (Seo et al., 2009). Also, temporary signal loss would be a significant continuity threat to aviation users.

In general, commercial GNSS receivers declare the occurrence of loss of lock based on C/N_0 or lock detector, but their criteria are dependent on the receiver design or user settings (Forte, 2012). In this study, the loss of lock is declared when the signal fades below a certain threshold of C/N_0 for a given baseline C/N_0 . For example, loss of lock is declared to occur at signal fading of -10 dB when the C/N_0 threshold is 30 dB-Hz for a baseline C/N_0 of 40 dB-Hz.

The impact of loss of lock is quantified with the time fraction of L1/L5 concurrent deep fading (where the threshold of fading depth is determined with a baseline C/N_0 and a C/N_0 threshold) for each scintillation simulation. The measure potentially leads to concurrent signal losses on both L1/L5 dual-frequency signals (i.e., complete loss of satellite) which may directly degrade the aviation availability. The corresponding fading processes in L1/L5 dual-frequency signals are modeled using the Markov chain-based method proposed in Sun et al. (2021) for the simulation of the occurrence of loss of lock in the availability analysis.

4. EXHAUSTIVE SEARCH FOR THE WORST SCINTILLATION EFFECTS

4.1. Scintillation indicators: S_4 and τ_0

Amplitude scintillation index S_4 and decorrelation time τ_0 are the most widely used scintillation indicators to represent variations in signal intensity fluctuations and their temporal variability, respectively. In particular, S_4 has been proposed as a discriminator or monitoring statistic to screen out satellite measurements affected by scintillations for GNSS aviation users by taking advantage of its near real-time availability (Saito et al., 2019; Scheitlin et al., 2015).

Xu et al. (2020) developed a scintillation simulator that only requires S_4 and τ_0 to represent typical equatorial scintillation scenarios based on the relationship between the indicators $\{S_4, \tau_0\}$ and the phase screen parameters $\{U, \rho_F/v_{eff}\}$, while spectral parameters $\{p_1, p_2, \mu_0\}$ are fixed with their typical values for equatorial scintillations. However, the numerical mapping between $\{S_4, \tau_0\}$ and the phase screen parameters may not cover all possible scintillations, especially for the cases with spectral parameters far apart from their typical values. Furthermore, S_4 values typically calculated with a time window of 60 s may induce significant statistical uncertainties because fluctuations of lower frequencies (with time scales much longer than 60 s) may not be sampled enough to be statistically significant with a time interval of 60 s, and similar problems may arise for τ_0 which uses a time window of 60 s as well (Sun et al., 2023).

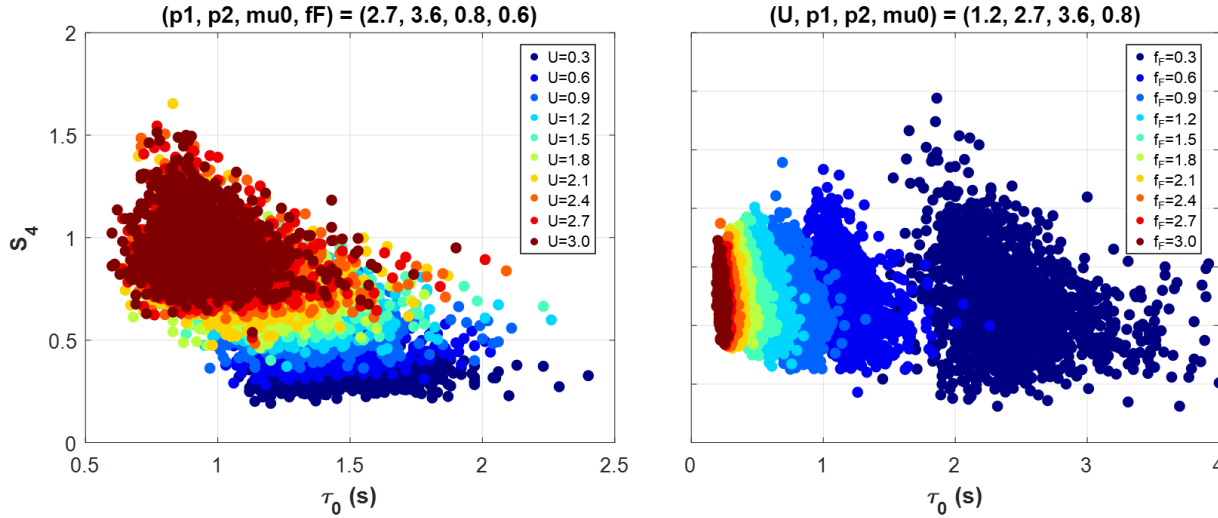


Figure 2 Distributions of $\{S_4, \tau_0\}$ pairs calculated with a time window of 60 s for scintillation simulation with $\{p_1, p_2, \mu_0, f_F\} = \{2.7, 3.6, 0.8, 0.6\}$ according to U (left) and $\{U, p_1, p_2, \mu_0\} = \{1.2, 2.7, 3.6, 0.8\}$ according to f_F (right)

Figure 2 shows an example of $\{S_4, \tau_0\}$ pairs with a 60-s cadence calculated from scintillation simulation with a given set of phase screen parameters. The left panel of Figure 5 shows the distribution of $\{S_4, \tau_0\}$ pairs according to U , and the right panel shows the distribution according to f_F . The $\{S_4, \tau_0\}$ values calculated with a time window of 60 s show large variability for the same set of phase screen parameters, in particular, the $\{S_4, \tau_0\}$ values get widely distributed as f_F decreases, as suggested in Sun et al. (2023).

To utilize S_4 and τ_0 as indicators to represent the scintillation impacts on availability of GNSS-based augmentation systems, we need to consider the uncertainties in the S_4 and τ_0 calculations. In other words, for each $\{S_4, \tau_0\}$ pair, we need to cover all possible scintillation scenarios specified by a parameter set $\{U, p_1, p_2, \mu_0, f_F, C/N_0\}$ that possibly result in the given $\{S_4, \tau_0\}$.

4.2. The Worst Scintillation Impacts according to S_4 and τ_0

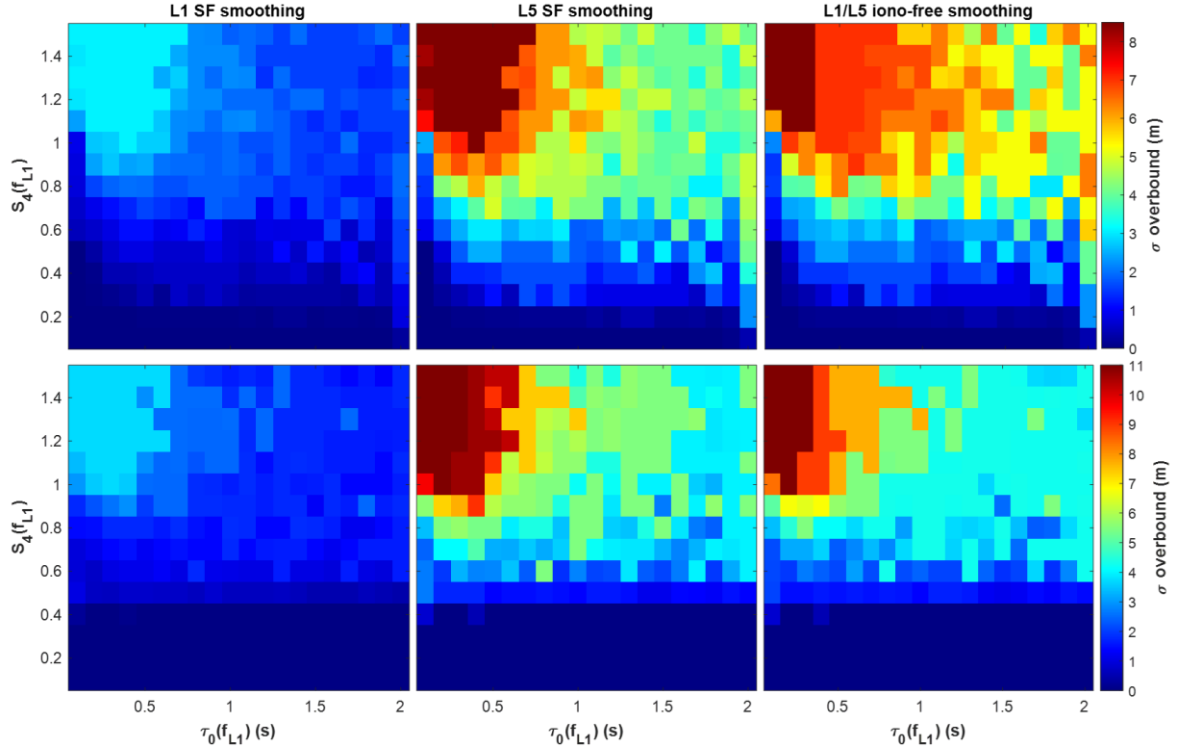


Figure 3 The worst σ_{scin} overbound results for L1, L5 single-frequency and L1/L5 ionosphere-free smoothing modes according to S_4 and τ_0 . The upper panel shows the cases with a baseline C/N_0 of 50 dB-Hz and the lower panel shows the cases with a baseline C/N_0 of 30 dB-Hz

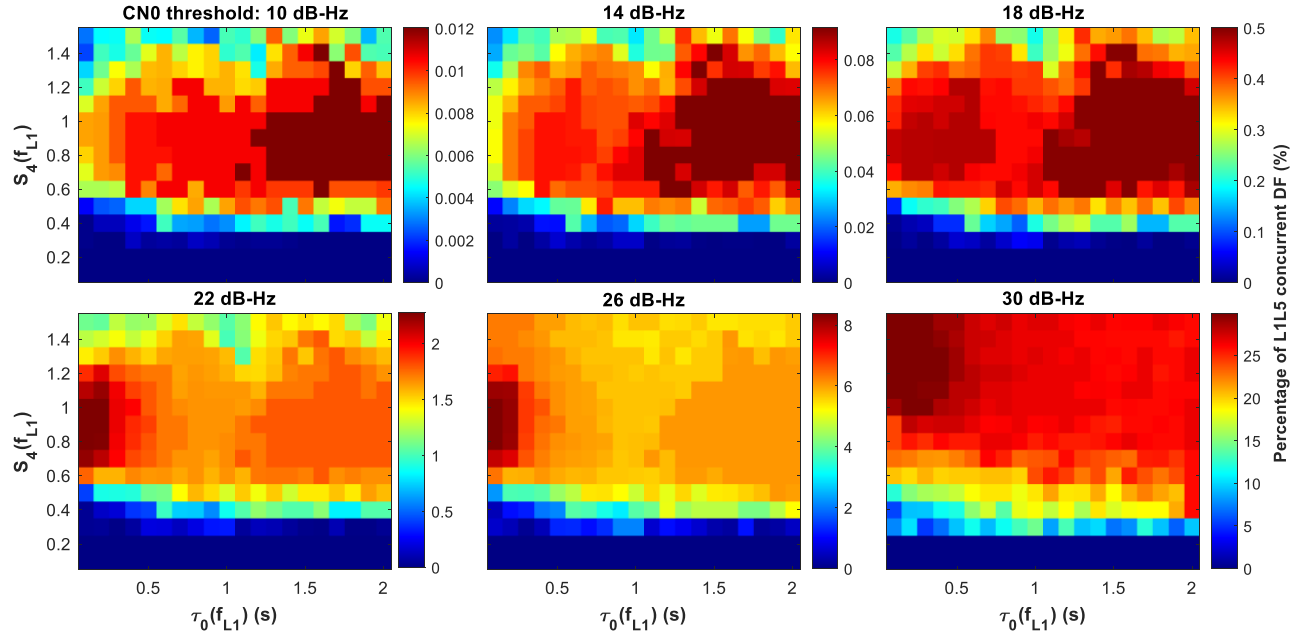


Figure 4 Time percentage of L1/L5 concurrent deep fading according to S_4 and τ_0 with different C/N_0 threshold for a baseline C/N_0 of 35 dB-Hz.

Figure 3 shows the worst scintillation impacts due to phase transitions (i.e., σ_{scin} overbound of L1, L5 single frequency and ionosphere-free mode) according to scintillation indicators S_4 and τ_0 . The upper panel shows the results with a baseline C/N_0 of 50 dB-Hz and the lower panel shows the results with a baseline C/N_0 of 30 dB-Hz. The σ_{scin} overbound values for a different baseline C/N_0 show similar trends according to S_4 and τ_0 , while large σ_{scin} overbound are obtained for a lower baseline C/N_0 which may be caused by the more frequent occurrence of noise-induced phase transitions with a higher noise level. For all smoothing modes, the larger σ_{scin} overbound is obtained at a higher S_4 and a lower τ_0 where the most deep and frequent signal fades are expected. The comparable σ_{scin} overbound values for L5 single-frequency and L1/L5 ionosphere-free smoothing mode indicate that the phase transition effects on the ionosphere-free combination are dominantly contributed from the L5 signal rather than the L1 signal.

Figure 4 shows the worst scintillation impacts due to loss of lock (i.e., time percentage of L1/L5 concurrent deep fading) according to S_4 and τ_0 with a different C/N_0 threshold for a baseline C/N_0 of 35 dB-Hz. The lower C/N_0 threshold designates the deeper signal fades while the higher C/N_0 threshold designates the shallower signal fades. In the case of a C/N_0 threshold of 10 dB-Hz, the overall percentage of L1/L5 concurrent deep fading is very low because a very deep fading level is required for the occurrence of loss of lock (-25 dB). In the case of a C/N_0 threshold of 30 dB-Hz, the overall percentage of concurrent deep fading is relatively high because of very low fading level for the occurrence of loss of lock (-5 dB). The highest rate of L1/L5 concurrent deep fading at S_4 above 0.8 and τ_0 below 0.5 may be due to very deep and frequent fades for a depth of only -5 dB. The time percentage of L1/L5 concurrent fading is only used for the search for the worst scintillation impacts.

5. AVAILABILITY SIMULATION

5.1. Simulation Specification

This study assumes the use of dual-constellation (GPS and Galileo), dual-frequency (L1/E1 and L5/E5a) advanced receiver autonomous integrity monitoring (ARAIM) that supports the localizer performance with vertical guidance (LPV)-200 requiring a high level of integrity, continuity, and availability. The Matlab Algorithm Availability Simulation Tool (MAAST) (Jan et al., 2009) is used to calculate availability as the percentage of operation time for which the vertical protection level (VPL) is lower than the vertical alert limit (VAL) of 35 m for the LPV-200 operation. VPLs are calculated based on the broadcast integrity parameters and error models which provide uncertainty for each error source. A 1 m of user range accuracy (URA) σ_{URA} is assumed for clock and ephemeris errors, and the Wide Area Augmentation System (WAAS) MOPS troposphere error model is used for tropospheric errors σ_{tropo} (RTCA, 2020). We use the Local Area Augmentation System (LAAS) MOPS airborne accuracy designator (AAD)-A and airframe multipath designator (AMD)-A model for single-frequency or dual-frequency ionosphere-free combination mode of code noise and multipath errors σ_{user} . Then the VPL is calculated every second based on the sum of all error components for the i^{th} satellite as follows,

$$\sigma_i^2 = \sigma_{URA,i}^2 + \sigma_{tropo,i}^2 + \sigma_{user,i}^2 \quad (19)$$

The loss of lock occurrences in L1/L5 signals are simulated by using the fading process model which corresponds to the set of simulation parameters $\{U, p_1, p_2, \mu_0, f_F, C/N_0\}$ for the highest rate of concurrent fading at a given S_4 , τ_0 , and a baseline C/N_0 . The fading process model generates a fading process with a 0.01 s time interval where the fading depth is enough to cause loss of lock for a given C/N_0 threshold.

The default mode is the ionosphere-free mode of dual-frequency signals; however, we consider temporary single-frequency operations when one frequency signal is temporarily lost. In this case, we use the single-frequency smoothing mode for σ_{user} and use the WAAS MOPS ionospheric correction model to additionally consider the user ionospheric range error (UIRE) σ_{UIRE} which is the residual error term after the ionospheric correction with Klobuchar model (RTCA, 2020). If both L1 and L5 signals are lost in tracking, then we consider a complete satellite loss by excluding the satellite from VPL calculations.

We assume that the carrier smoothing filter is initialized when there is at least one occurrence of loss of lock at L1 or L5 within a time interval (1 s) for VPL calculations. As specified in DFMC SBAS MOPS (EUROCAE, 2021), we apply the transient phase of the multipath/code noise error model (σ_{user}) when the elapsed time after the smoothing filter re-initialization is less than 360 s for all smoothing modes. Note that we assume that single-frequency and ionosphere-free smoothing can be implemented in parallel.

We incorporate the effects of phase transitions in VPL calculations by using the worst σ_{scin} overbound value for a given S_4 , τ_0 , and a baseline C/N_0 as follow:

$$\sigma_i^2 = \sigma_{\text{URA},i}^2 + \sigma_{\text{tropo},i}^2 + \sigma_{\text{user},i}^2 + \sigma_{\text{scin},i}^2(S_4, \tau_0, C/N_0) \quad (20)$$

We use σ_{scin} of L1/L5 ionosphere-free smoothing mode during dual-frequency tracking, but we use σ_{scin} of L1-only or L5-only smoothing mode for a temporary single-frequency mode when a signal on one frequency is briefly lost.

5.2. Scintillation Scenario

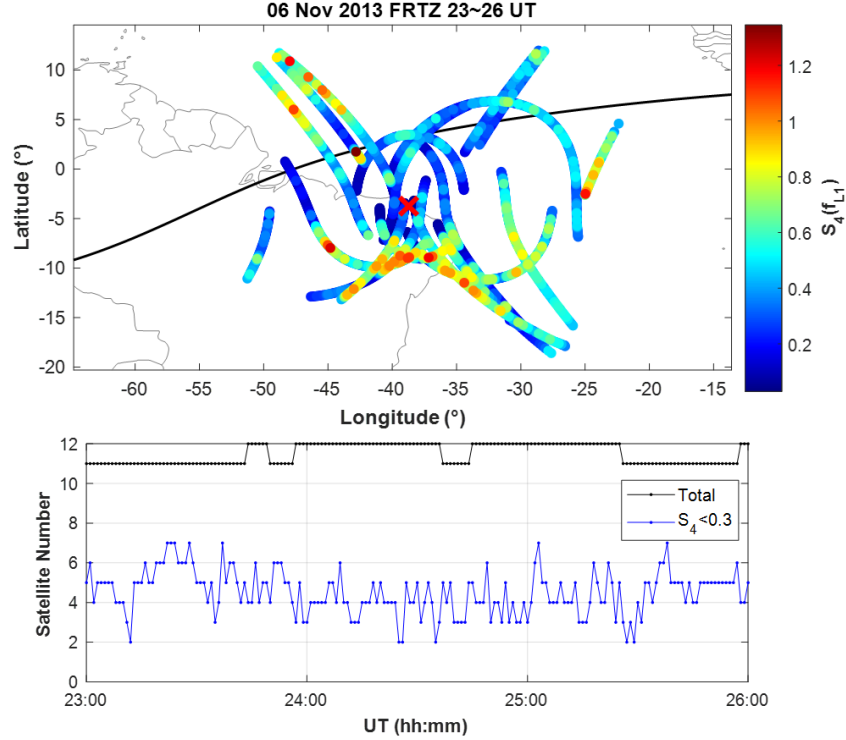


Figure 5 The 3-hour scintillation data observed at FRTZ (3.74°S, 38.72°W) from 23 UT on 06 Nov 2013 to 02 UT on 07 Nov 2013. The upper panel shows the geographic distribution of S_4 of L1, and the lower panel shows the number of available satellites and non-scintillating satellites.

The scintillation data were collected at five stations, FRTZ (3.73°S, 38.72°W), PALM (10.20°S, 48.31°W), POAL (30.07°S, 51.12°W), PRU2 (22.12°S, 51.41°W), and SJCE (23.21°S, 45.86°W) in Brazil, from November 2013 to January 2014 (near the maximum phase of solar cycle 24). The 50-Hz multi-constellation multi-frequency GNSS measurements were collected using Septentrio PolaRxS Pro scintillation receivers with a visibility mask angle of 5°.

The most severe scintillation data was selected based on the maximum S_4 of L1 and the number of scintillating satellites (among GPS and Galileo constellations). The selected data was collected in FRTZ station from 23 UT on 06 Nov 2013 to 02 UT on 07 Nov 2013, where the maximum S_4 of L1 reaches 1.4 and the maximum 10 out of 12 satellites are under scintillation ($S_4(f_{L1}) > 0.3$) as shown in Figure 5. We use time series of S_4 , τ_0 , and baseline C/N_0 of L1/E1 for all visible GPS and Galileo satellites by mapping into the corresponding scintillation impacts presented in Section 4.2 for availability simulations.

5.3. Availability Results

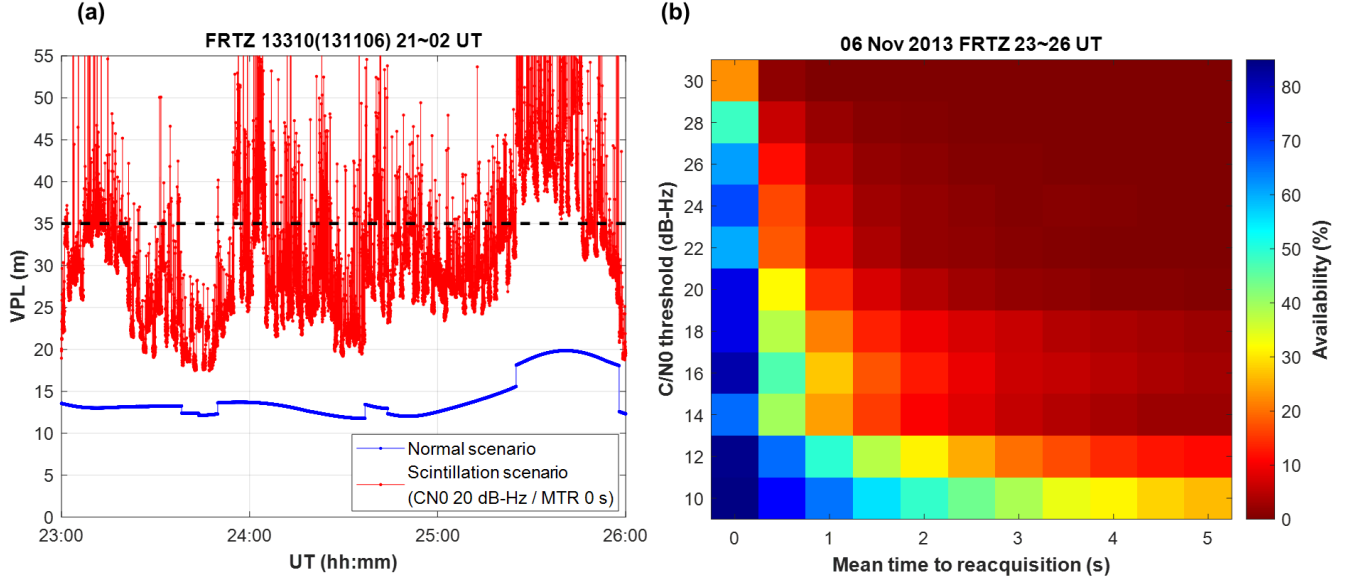


Figure 6 (a) Simulated VPL with and without considering scintillation effects. (b) LPV-200 availability results according to mean time to reacquisition and C/N_0 threshold

Figure 6(a) shows the simulated VPL for the selected scintillation scenario presented in Section 5.2. The VPL of the nominal case (blue line) was obtained with the actual satellite geometry of the period but the scintillation effects are not considered. The VPL in the best case without any scintillation effects is always below the 35 m VPL for LPV-200 approach, that is, 100% availability. However, VPL significantly increases when the scintillation effects are incorporated in the simulation (red line). The scintillation case in Figure 6(a) is simulated with a C/N_0 threshold of 20 dB-Hz and a reacquisition time of 0 s. Note that this case doesn't account for the effects of signal loss due to a 0-s reacquisition time and only accounts for the effects of phase transitions (σ_{scin}) and smoothing filter reset.

Figure 6(b) shows the availability results for LPV-200 operation under the selected scintillation scenario with varying mean time to reacquisition and C/N_0 threshold. The simulations were performed parametrically based on the two parameters which are dependent on the receiver settings and the reacquisition capability. The availability contour plots show that the availability is very sensitive to both parameters: mean time to reacquisition and C/N_0 threshold. For example, availability with a C/N_0 threshold of 30 dB-Hz is about 23% even with immediate reacquisition of signal tracking. Also, availability with a 5-s reacquisition time is about 26% even with a C/N_0 threshold of 10 dB-Hz.

6. CONCLUSION

This study proposed a new approach to conservatively quantify scintillation impacts in terms of GNSS-based augmentation systems. We investigated a wide range of equatorial scintillation by simulating all permutations of the phase screen parameters $\{U, p_1, p_2, \mu_0, f_F\}$ and a baseline C/N_0 within their empirical range. The scintillation effects for each simulation of a given set of parameters were conservatively quantified by using overbounding statistics of smoothing noise level for phase transitions and the percentage of dual-frequency concurrent fading (with its fading depth enough to cause loss of lock for a given C/N_0 threshold).

We used near real-time scintillation indicators, the amplitude scintillation index S_4 and the decorrelation time τ_0 , to specify equatorial scintillation conditions for the availability assessment. Based on the S_4 and τ_0 values with a 60-s window calculated from scintillation realizations, we assigned the scintillation effects to $\{S_4, \tau_0\}$ grids and searched for the worst-case scintillation impacts for each set of $\{S_4, \tau_0, C/N_0\}$. The LPV-200 availability results of GPS/Galileo dual-frequency ARAIM for the selected scintillation scenario specified by $\{S_4, \tau_0\}$ show that the availability is highly dependent on the reacquisition time and the C/N_0 threshold which are closely related to the tracking capability and the user settings of aviation receivers.

The proposed approach allows availability assessment of GNSS-based augmentation systems for any scintillation conditions specified by indicators $\{S_4, \tau_0\}$ where the indicators can be obtained in near real-time. This work can be utilized to provide performance standards and mitigation strategies for GNSS-based augmentation systems under various scintillation conditions. Furthermore, this can be utilized in the design of scintillation monitors for such systems which can calculate $\{S_4, \tau_0\}$ from receiving measurements in near real-time.

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